The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5306
Subject Title	Electronics Design and Informatics for Low-altitude Economy
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will provide students with: 1. Provide students with advanced knowledge and skills in information and electronics system design for low-altitude vehicles, including general concepts of circuit design, embedded systems, power management, and sensor integration for drones. 2. Enable students to analyze, design, and optimize information and electronic hardware components and architectures for unmanned aerial systems (UAS) and other low-altitude platforms, considering factors such as power efficiency, weight constraints, and reliability. 3. Equip students with the ability to develop and integrate advanced information and electronic subsystems, such as flight navigation, sensor integration and controllers, communication modules, and payload interfaces, for enhanced functionality and performance in low-altitude applications. 4. Prepare students to address the unique challenges of information and electronics design in the low-altitude domain, such as harsh environmental conditions, electromagnetic interference, and regulatory compliance.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Apply advanced concepts and techniques in electronics design and informatics to develop, operate, maintain, and diagnose sophisticated low-altitude systems. b. Design and optimize information and electronic hardware components, architectures, and embedded software for UAS and other low-altitude platforms, considering factors such as power efficiency, weight constraints, and reliability. c. Develop and integrate advanced electronic subsystems, such as flight navigation, sensor integration and controllers, communication modules, and payload interfaces, for enhanced functionality and performance. d. Analyse and address the unique challenges of electronics design and informatics in the low-altitude domain, such as harsh environmental conditions, electromagnetic interference, and regulatory compliance. e. Innovate and create new solutions, algorithms, and architectures that advance the state-of-the-art in low-altitude electronics and informatics.

Subject Synopsis/ Indicative Syllabus	**1. Advanced Electronics and System Design for Low-Altitude Systems** • Circuit design and optimization for low-altitude vehicles • Embedded systems and microcontroller programming for UAS • Power management and energy efficiency in low-altitude electronics **2. Information Technologies for Low-Altitude Systems** • Data processing, storage, and transmission in low-altitude systems • Cybersecurity and data protection in low-altitude applications • Cloud computing and edge computing for low-altitude informatics • Sensor principles and mathematical models: GPS/GNSS, LiDAR, cameras, and IMU • Statistical analysis and mathematical foundations of sensor fusion • Error modeling, uncertainty propagation, and stochastic processes • LiDAR SLAM algorithms and point cloud registration techniques **3. Practical Implementation and Applications** • Data collection methodologies and statistical analysis techniques • Non-linear optimization theory and numerical methods • Implementation of SLAM/Navigation algorithm • Practical sensor integration and system validation • Electronic system design for harsh environmental conditions • Regulatory compliance and certification for low-altitude systems • Robot Operating System (ROS) architecture and implementation **4. Challenges and Innovations in Low-Altitude Electronics and Informatics** • Harsh environmental conditions and ruggedization • Electromagnetic interference and compatibility • Regulatory compliance and certification • Emerging technologies and future trends in low-altitude electronics and informatics
Teaching/Learning Methodology	1. The teaching and learning methods include lectures sessions, assignments, course project and examination. 2. The continuous assessment and examination are aimed at providing students with integrated knowledge required for electronics design for UAV. 3. Technical/practical examples and problems are raised and discussed in class sessions.

Teaching/Learning Methodology	Outcomes				
	a	b	c	d	e
Lecture	√	√	√	√	√
Assignment	√	√	√	√	√
Course Project			√	√	√

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
		a	b	c	d	e
1. Assignment	20%	√	√	√	√	√
2. Midterm Test	10%	√	√	√		
2. Course Project	20%			√	√	√
3. Final examination	50%	√	√	√	√	√
Total	100%					

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Overall Assessment:

$$0.5 \times \text{Continuous Assessment} + 0.5 \times \text{Final Examination}$$

The continuous assessment (50%) is aimed at enhancing the students' comprehension and assimilation of various topics of the syllabus via assignments and a course project. The final examination (50%) will also be considered to assess the students' learning outcomes.

Student Study Effort Expected

Class contact:	
▪ Lecture/Case Study	39 Hrs.
Other student study effort:	
▪ Self-learning/preparation	66 Hrs.
Total student study effort	105 Hrs.

Reading List and References

[1] P. Horowitz and W. Hill, *The Art of Electronics*, 3rd ed. Cambridge University Press, 2015. ISBN: 978-0-521-80926-9.

[2] K. P. Valavanis and G. J. Vachtsevanos, Eds., *Handbook of Unmanned Aerial Vehicles*. Springer, 2015. doi: 10.1007/978-90-481-9707-1.

[3] J. Gundlach, *Designing Unmanned Aircraft Systems: A Comprehensive Approach*. American Institute of Aeronautics and Astronautics, 2012. doi: 10.2514/4.869204.

[4] Q. Quan, *Introduction to Multicopter Design and Control*. Springer, 2017.

[5] M. Mozaffari, W. Saad, M. Bennis, Y. Nam, and M. Debbah, "A Tutorial on UAVs for Wireless Networks: Applications, Challenges, and Open Problems," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 3, pp. 2334-2360, 2019. doi: 10.1109/COMST.2019.2902862.

May 2025